

Emotional tone moderation: what the experiments measured

Every figure below comes from a run recorded in the project database and judged by Claude Opus 5 blind, one story per call. Where a number carries a caveat, the caveat is next to it.

1. The question, and the instrument

A general LLM writes a data story from an evidence pack; a second, agentic LLM moderates its emotional tone; a separate pass fact-checks the result. The claim under test is that moderation makes the tone more like a person's without costing accuracy.

Tone is scored on two axes, each 1-5, where both ends are failures and 3 is calibrated. Alarmism runs from flat to catastrophising; optimism from bleak to false reassurance. Two axes rather than one because the datasets fail in opposite directions: a falling series tempts glossing over remaining harm, which a single alarmism axis scores as a calm 2.0 while it is exactly as miscalibrated.

The judge is **Claude Opus 5**, called through the CLI outside the Ollama stack. It never generates and never moderates, so it grades no work of its own, and it shares no vendor or family with the models under test. Each story is scored in its own call, with no label and no sibling to compare against, so the judge never learns which story was the treatment.

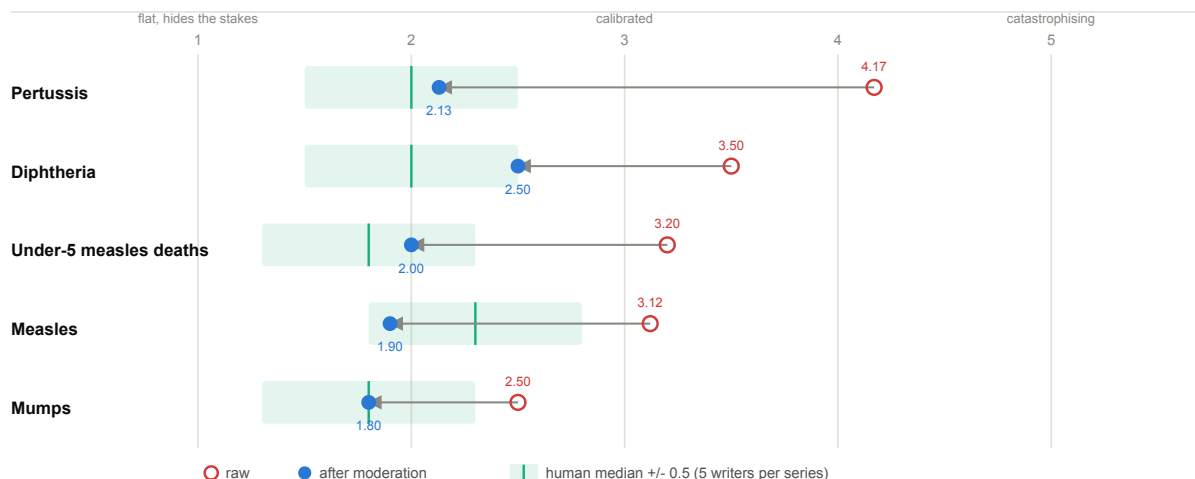
2. Does moderation reach the human level?

	RAW	MODERATED	HUMAN
alarmism	3.74	2.09	2.00
optimism	2.15	2.60	2.50

The two axes move in opposite directions and both land within 0.2 of the team's own writers, 25 stories across 5 series. Alarmism falls, optimism rises, both toward the middle.

Moderation moves machine stories onto the human level

Alarmism 1-5, Claude Opus judging blind, one story at a time. 32 runs against 25 human stories. Overall 3.74 -> 2.09, human median 2.00.



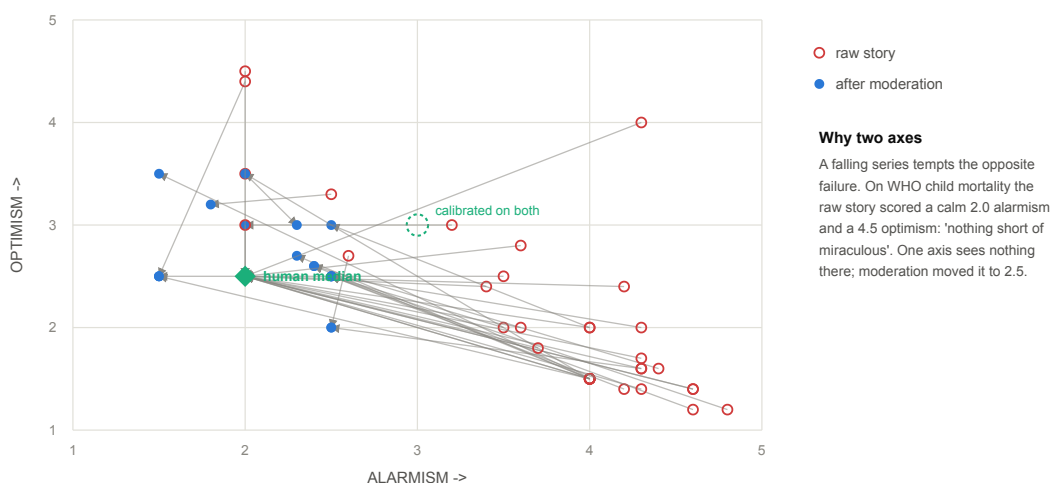
The runs are not spread evenly: pertussis carries 24 and measles carries 5, and the remaining series are a single run each, so those arrows are one story rather than a mean. The human stories carry no headline and the machine stories do; headlines concentrate alarmism, so the human line is if anything flattered and every gap shown is a lower bound.

Every series moves toward the human band. Pertussis, the hottest, travels furthest.

Caveat. The human stories carry no headline and the machine stories do. Headlines are where alarmism concentrates, so the human column is if anything flattered and every gap shown is a lower bound. The set is also not the four-named-writer shape the protocol specifies, so nothing here computes H.

Two axes, because a falling series fails in the other direction

Every run: 35 raw-to-moderated moves, all eight datasets.



Alarmism falls 3.74 -> 2.09 and optimism rises 2.15 -> 2.60, both toward the middle. Note the moderated cloud settles below alarmism 3: the moderator aims at quiet rather than at calibrated, and lands near where these human writers also sit (diamond).

Why two axes. On WHO child mortality the raw story scored a calm 2.0 for alarmism and 4.5 for optimism; a single-axis judge sees no problem there.

3. Which generator and moderator

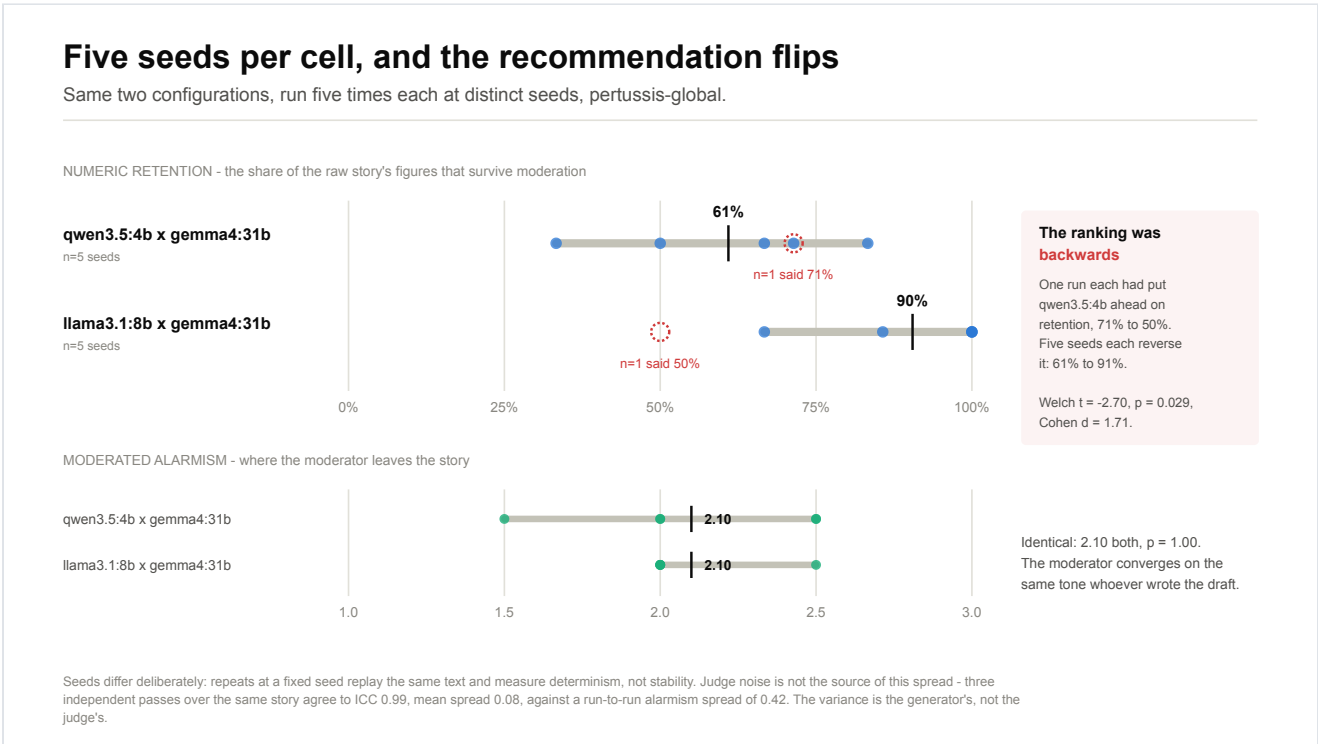
Thirteen combinations were run on pertussis-global, one each. Six of them land on the human median of 2.0 exactly, so tone does not discriminate between them and faithfulness has to. That single-run table is reproduced as fig10 in the appendix; the conclusion below supersedes it.

Five seeds per cell changed the answer

CONFIG	MODERATED ALARMISM	NUMERIC RETENTION
qwen3.5:4b × gemma4:31b	2.10 ± 0.42	61% ± 20%
llama3.1:8b × gemma4:31b	2.10 ± 0.22	90% ± 15%

The single-run ranking was backwards. One run each had put qwen3.5:4b ahead on retention, 71% to 50%. Five distinct seeds each reverse it: 61% against 90%, Welch t = -2.7, p = 0.0288, Cohen d = 1.71. Both original values were unrepresentative draws from distributions spanning 33-83% and 67-100%.

Moderated alarmism is identical, 2.10 both, p = 1.00. So the generator decides what survives moderation, not where the tone lands. Recommendation: **llama3.1:8b × gemma4:31b**.



The flip, with the test on the figure.

4. How much can the instrument be trusted?

Every story was rated three times, 192 calls in total.

MEASURE	ALARMISM	OPTIMISM
ICC(2,1), absolute agreement	0.991	0.983
Krippendorff's alpha	0.991	0.982
mean spread over three passes	0.08	0.10

41 of 64 stories scored identically on all three passes and 64 of 64 fell within 0.5. The widest disagreement on any single story was 0.5 (pertussis-global/g3b, moderated).

This decomposes the variance in section 3. The judge disagrees with itself by 0.08 points; the same configuration re-seeded disagrees with itself by 0.42. The spread in the repeat experiment is the generator resampling, not the instrument wobbling.

Same model and same prompt on all three passes, so this is *self-consistency*, not inter-rater reliability. It is an upper bound on what a different judge would agree to.

The tone scale now has an error bar

Every story rated three times by Claude Opus, 192 calls, \$3.25.

ICC(2,1)

0.991

two-way random, absolute agreement

Krippendorff alpha

0.991

interval scale

mean spread

0.08

points, over three passes

max spread

0.5

worst single story

Agreement across all 64 stories, three independent passes each



Why this matters for every other number in the deck

The judge disagrees with itself by 0.08 points on average. The same configuration run at five different seeds disagrees with itself by 0.42. So the spread in the repeat experiment is the generator resampling, not the instrument wobbling - which is what makes it worth measuring rather than averaging away.

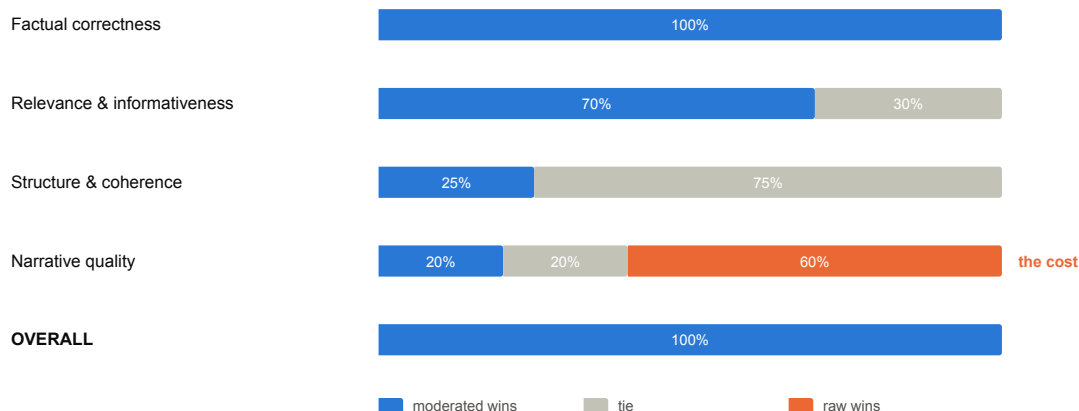
Same model and same prompt on all three passes, so this is self-consistency, not inter-rater reliability: it is an upper bound on what a different judge would agree to. Calls are stateless and share no context, which is the only sense in which they are independent.

5. Does the reader get a better story?

The published pairwise protocol, with two departures: position is randomised per pair, and the judge is never told which story was moderated.

The moderated story wins on everything except readability

20 blinded pairs, position randomised, Claude Opus deciding each criterion.



The judge is never told which story was moderated, and which one is shown first is decided by a coin flip per pair. Narrative quality is the one criterion the raw story wins, and it wins it three times as often: this system buys calibration and accuracy by spending engagement, and a deck that hides that row is selling rather than reporting.

Moderated wins overall and on factual correctness; it loses narrative quality.

Narrative quality is the cost. It is the only criterion the raw story wins, and it wins it three times as often. The system buys calibration and accuracy by spending engagement.

6. Model scale versus model family

Within the qwen3.5 family, raw alarmism does not fall with size: 4.8 at 2B, 4.6 at 4B, 4.5 at 9B and 4.3 at 27B. The one calm generator in the whole set is `llama3.1:8b` at 3.98 over five seeds.

Scaling a qwen by more than 13× buys none of that calm, so **writing calmly is a family trait here, not a size effect**. An earlier reading of the ladder as "smaller generators write hotter copy" was a family effect misread as a scale one; the 27B rung is what settles it.

7. What these numbers cannot support

- **n = 1 outside the two repeated cells.** Every other combination is a single run, and section 3 is a worked demonstration of how badly that can mislead.
- **One judge.** Section 4 measures self-consistency, not agreement between independent raters. A second judge model is the missing control.
- **Breadth and depth do not intersect.** Only one tier spans all seven datasets; the rest ran on pertussis alone.
- **The grid is L-shaped, not factorial.** Generators were varied against one moderator and moderators against one generator, so the best × best cell was never run.
- **Effective sample size is smaller than it looks.** Five of the seven datasets are WHO surveillance series of the same shape, and two are nested, so they are closer to three independent series than

seven.

Appendix: the single-run combination table

Six combinations tie on tone. Faithfulness breaks the tie.

Superseded by fig13: five seeds per cell reverse the retention column. Read this as the ranking it is, not as a recommendation.

generator x moderator	alarmism	gap to human	figures kept	invented	chrF++ to human	seconds
llama3.1:8b x gemma4:31b	2.0	0.0	50%	0	0.394	268
llama3.2:1b x gemma4:31b	2.0	0.0	33%	0	0.356	221
qwen3.5:4b x gemma4:31b	2.0	0.0	71%	0	0.285	194
llama3.1:8b x gemma4:26b	2.0	0.0	33%	0	0.376	96
llama3.1:8b x qwen3.5:9b	2.0	0.0	33%	1 invented	0.389	93
qwen3.5:4b x gemma4:31b	2.0	0.0	71%	0	0.285	230
llama3.1:8b x gemma4:31b	2.0	0.0	100%	0	0.381	327
qwen3.5:27b x gemma4:31b	2.0	0.0	75%	0	0.365	281
llama3.1:8b x gemma4:12b	2.4	0.4	50%	0	0.408	315
qwen3.5:9b x gemma4:31b	2.4	0.4	50%	0	0.382	232
llama3.1:8b x gemma4:31b	2.5	0.5	75%	0	0.423	291
llama3.2:3b x gemma4:31b	2.5	0.5	20%	0	0.403	263
llama3.1:8b x qwen3.6:35b	1.5	0.5	33%	1 invented	0.339	89
qwen3.5:2b x gemma4:31b	2.5	0.5	29%	0	0.407	264

Pertussis only, one row per run. Where a pair of models appears twice or more, those are separate runs of the same configuration and "figures kept" disagrees with itself: 50%, 75% and 100% for llama3.1:8b x gemma4:31b. That spread is why one run cannot rank anything, and it is what fig13 measures. The grid is L-shaped, so the best x best cell was never run. Red rows invented a figure.

Superseded by section 3. Kept because it is what the project believed before the cells were repeated.